Classical Machine Learning for Automated Sickle Cell Detection from Peripheral Blood Smear Images: A Ugandan Clinical Dataset Study

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***Abstract*—Sickle cell disease (SCD) affects approximately 13.3% of the Ugandan population, yet diagnosis remains inaccessible in rural settings due to the requirement for trained laboratory personnel and specialist equipment. This study applies classical machine learning to automate sickle cell detection from peripheral blood smear images using the Tushabe et al. (2024) Ugandan clinical mobile-phone microscopy dataset — the first study to apply machine learning to this dataset. Histogram of Oriented Gradients (HOG) feature extraction converts each image into a 8100-dimensional numerical feature vector. Two classifiers are trained and compared: Support Vector Machine (SVM) and Random Forest. Both models are optimised using GridSearchCV with 5-fold cross-validation. The SVM achieves the best overall performance with 91.98% accuracy and AUC-ROC of 0.9651, while Random Forest achieves higher sensitivity of 96.47%. Both models show overfitting on the small dataset, which is identified and discussed. Results demonstrate that classical machine learning with HOG features can produce meaningful results for sickle cell detection as a low-resource screening tool.**

***Index Terms*—Sickle Cell Disease, Machine Learning, HOG Features, SVM, Random Forest, Blood Smear Classification, Uganda, Low-Resource Healthcare**

I. Introduction

Sickle cell disease is a genetic blood disorder caused by a mutation in the HBB gene that produces haemoglobin S, causing red blood cells to deform into a crescent shape under deoxygenated conditions. Uganda carries a high disease burden, with the sickle cell trait estimated at 13.3% nationally and reaching 19.8% in Kampala [1]. An estimated 15,000 to 20,000 children are born with SCD in Uganda each year, many of whom go undiagnosed due to limited access to specialist diagnostic facilities in rural areas.

Current diagnosis depends on laboratory-based peripheral blood smear microscopy, haemoglobin electrophoresis, and sickling tests, all of which require trained personnel and equipment unavailable in most Ugandan district hospitals. Machine learning offers the possibility of automating diagnosis from blood smear images captured using mobile phone cameras on basic microscopes, making screening accessible to non-specialist health workers in rural settings.

This study makes two specific contributions. First, it is the first machine learning study applied to the Tushabe et al. (2024) Ugandan clinical microscopy dataset. Second, it provides a systematic comparison of SVM and Random Forest classifiers using HOG features, with full overfitting analysis and hyperparameter optimisation.

II. Related Work

Machine learning has been applied to blood cell analysis with strong results. Acevedo et al. [2] published a dataset of 17,092 annotated peripheral blood cell images and demonstrated automated classification across eight cell types. Deep learning approaches, particularly convolutional neural networks, have achieved high accuracy in medical image classification tasks including diabetic retinopathy, chest X-ray analysis, and blood smear interpretation [3].

For sickle cell detection specifically, prior studies have used datasets collected under controlled laboratory conditions in high-income countries. None have used mobile-phone microscopy data from clinical settings in sub-Saharan Africa. Poostchi et al. [4] reviewed machine learning approaches for blood smear analysis and established the feasibility of mobile-phone microscopy as an imaging platform, but did not address sickle cell detection in Ugandan clinical contexts.

HOG features have been widely used for object and shape recognition since Dalal and Triggs [5] demonstrated their effectiveness for pedestrian detection. Their ability to capture gradient orientations makes them well suited for capturing the morphological differences between sickle and normal red blood cells.

III. Dataset

A. Data Sources

Two datasets were combined. The primary dataset is the Tushabe et al. (2024) Ugandan Sickle Cell Microscopy Dataset [1], containing blood smear images collected from patients in Soroti and Kumi districts, Uganda, using mobile phone cameras placed on basic optical microscopes. The dataset includes 422 unlabelled sickle cell positive images and 147 normal images. An additional 422 labelled images were excluded because bounding box annotations were found burned directly into the image pixels, which would cause a classifier to learn annotation artefacts rather than cell morphology.

The supplementary dataset is the BCCD (Blood Cell Count and Detection) Dataset [6], providing 364 normal peripheral blood smear images captured under controlled laboratory conditions. These supplement the negative class.

B. Dataset Composition

The final combined dataset contains 933 images: 422 positive (45.2%) and 511 negative (54.8%), giving a mild imbalance ratio of 1.21:1. A stratified 80/20 split produced 746 training samples and 187 test samples with class proportions preserved.

C. Data Challenges

Two data challenges were encountered and resolved. The Mendeley Peripheral Blood Cell dataset [2], originally planned as the normal class source, was found to contain segmented white blood cells rather than red blood cell smear images and was replaced with the BCCD dataset. Additionally, a domain gap exists within the negative class: Tushabe images carry a circular microscope border while BCCD images are rectangular, introducing a visual inconsistency that represents a known limitation.

IV. Methodology

A. HOG Feature Extraction

Raw images cannot be fed directly to classical machine learning algorithms, which require a tabular features matrix. HOG feature extraction converts each image into a fixed-length numerical vector by computing gradient orientation histograms in localised image regions. All images were resized to 128×128 pixels before extraction to eliminate image dimension as a potential spurious feature. HOG parameters used were 9 orientations, 8×8 pixels per cell, 2×2 cells per block, and L2-Hys block normalisation. Each image produced a feature vector of 8100 values.

B. Preprocessing Pipeline

The train-test split was performed before any preprocessing to prevent data leakage [7]. A numeric preprocessing pipeline applied StandardScaler to standardise all features to zero mean and unit variance. This is particularly important for SVM, which is sensitive to feature scale. The scaler was fitted on training data only and applied to transform both training and test sets.

C. Classifiers

Two classifiers were trained and compared.

Support Vector Machine (SVM): SVM finds the optimal hyperplane separating the two classes with maximum margin. The RBF kernel was used with class_weight=balanced to address class imbalance. GridSearchCV optimised C (0.1, 1, 10, 100), kernel (rbf, linear), and gamma (scale, auto) using 5-fold cross-validation with F1 scoring.

Random Forest (RF): Random Forest builds multiple decision trees on random feature subsets and combines predictions by majority vote. The class_weight=balanced parameter was used. GridSearchCV optimised n_estimators (100, 200), max_depth (10, 20, None), min_samples_split (2, 5, 10), and min_samples_leaf (1, 2).

D. Evaluation Metrics

Models were evaluated using accuracy, AUC-ROC, F1 score, sensitivity, specificity, and precision. Sensitivity is the most clinically important metric — missing a sickle cell patient is more harmful than a false positive referral. Overfitting was assessed by comparing training and test accuracy and using 5-fold cross-validation.

V. Results

A. SVM Results

The baseline SVM achieved 89.84% test accuracy and AUC-ROC of 0.9450, with a training-test gap of 6.54% indicating mild overfitting. GridSearchCV identified the best parameters as C=10, gamma=scale, and RBF kernel. The optimised SVM improved test accuracy to 91.98% and AUC-ROC to 0.9651, with 15 errors on 187 test images.

B. Random Forest Results

The baseline Random Forest achieved 87.70% test accuracy and AUC-ROC of 0.9333, with a training-test gap of 12.30% indicating significant overfitting. GridSearchCV identified best parameters as max_depth=20, min_samples_split=5, min_samples_leaf=1, and n_estimators=200. The optimised model achieved 87.17% accuracy and AUC-ROC of 0.9396. The overfitting gap was not fully resolved, remaining at 12.83%.

C. Model Comparison

Table I summarises the performance of both optimised models on the 187-sample test set.

TABLE I
MODEL PERFORMANCE COMPARISON ON TEST SET (N=187)

Metric	SVM	Random Forest
Test Accuracy (%)	**91.98**	87.17
AUC-ROC	**0.9651**	0.9396
F1 Score	**0.9133**	0.8723
Sensitivity (%)	92.94	**96.47**
Specificity (%)	**91.18**	79.41
Precision (%)	**89.77**	79.61
Errors / 187	**15**	24
Overfitting Gap (%)	**8.02**	12.83

SVM outperforms Random Forest on accuracy, AUC-ROC, F1, specificity, and precision. Random Forest achieves higher sensitivity of 96.47%, missing only 3 sickle cell cases compared to 6 for SVM. In a clinical screening context where minimising false negatives is the priority, Random Forest's higher sensitivity is a meaningful advantage.

D. Overfitting Analysis

Both models show overfitting. SVM cross-validation mean was 85.12% with standard deviation of 0.80%, compared to test accuracy of 91.98%. Random Forest cross-validation mean was 85.12% with standard deviation of 1.15%, compared to test accuracy of 87.17%. The overfitting is attributed to the small dataset size of 933 images combined with the high dimensionality of 8100 HOG features.

VI. Discussion

SVM is the better overall classifier for this task, achieving higher accuracy, AUC-ROC, and specificity. The RBF kernel effectively captures the non-linear decision boundary between sickle and normal cell morphological features in the HOG feature space. The higher regularisation parameter C=10 allowed the model to fit a more complex boundary while maintaining reasonable generalisation.

Random Forest's superior sensitivity makes it more suitable for deployment as a first-line screening tool where missing a positive case is unacceptable. The persistent overfitting gap reflects the inherent challenge of training ensemble methods on small datasets with high-dimensional features — even with depth constraints, trees find ways to memorise training samples.

Both models demonstrate that classical machine learning with HOG features can classify blood smear images with clinically useful accuracy. This is significant because classical methods are more computationally efficient than deep learning and can run on standard hardware without GPU access, making them viable for deployment in low-resource Ugandan health facilities.

VII. Limitations

The dataset of 933 images is small, limiting model generalisation. The domain gap between Tushabe mobile-phone images and BCCD laboratory images within the negative class introduces visual inconsistency. Both models show overfitting that was not fully resolved through hyperparameter tuning. No external clinical validation was performed.

VIII. Conclusion

This study applied SVM and Random Forest classifiers with HOG feature extraction to the Tushabe (2024) Ugandan clinical microscopy dataset for automated sickle cell detection. SVM achieved 91.98% accuracy and AUC-ROC of 0.9651, while Random Forest achieved higher sensitivity of 96.47%. Both models demonstrate that classical machine learning can produce meaningful results for this task. Future work should address the overfitting through larger datasets, explore additional feature extraction methods, and conduct clinical validation in Ugandan health facilities.

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